

Joint Training with Machine-Generated Text Detection Can Distort Sarcasm Detection: A Bilingual Diagnosis of Class-Conditional Negative Transfer

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Abstract

Sarcasm detection requires inferring intent beyond literal wording. LLM-generated examples may expand sarcasm resources, but evaluating models on such data requires separating intended style from source-conditioned behavior. When sarcasm and source predictions are desired, a shared dual-head encoder can reuse one representation; this sharing may allow source cues learned for machine-generated text detection (MGTD) to influence sarcasm decisions. We harmonize Chinese and English resources, generate topic-matched AI text, and build DualTest for source $\times$ sarcasm analysis, using requested styles as AI-side sarcasm labels. In the plain configuration, the false-positive rate relative to requested-style labels on AI-generated texts prompted to be non-sarcastic is 67.0 percentage points higher under the evaluated joint-training procedure than under a same-architecture single-task control. Within AI-generated texts, relative ranking remains largely preserved, indicating a source-conditioned score shift. The matched plain dual-single contrast retains the same direction under language stratification and fixed-detector evaluation of a second generator’s outputs. Representation diagnostics support a source-correlated shortcut hypothesis. An exploratory analysis shows that predicted human-source probabilities do not consistently track source-label-blinded LLM ratings of human-likeness. These findings motivate source $\times$ class subgroup evaluation and external validation before detector probabilities are reused as quality scores or rewards.

1 Introduction

Sarcasm is a pragmatic phenomenon in which a speaker’s intended attitude may differ from an utterance’s literal meaning. Interpreting it relies on implicit contrast, communicative intent, and context, making it difficult to detect automatically. Detection errors can reverse sentiment pre-

dictions and disrupt opinion-mining or content-moderation systems (Tay et al., 2018; Vidgen et al., 2019). Annotation outcomes also vary with elicitation method and annotator perspective (Oprea and Magdy, 2020; Casola et al., 2024), while model behavior is sensitive to domain, style, and language (Jang and Frassinelli, 2024; Casola et al., 2024) (§2.1).

One potential way to expand sarcasm resources is to generate additional examples with LLMs. Their usefulness depends on whether they express the intended style without allowing source-specific regularities to become shortcuts. Evaluation must therefore separate sarcasm behavior by source: human-authored and AI-generated texts can each be sarcastic or non-sarcastic, yielding all four combinations.

Source and sarcasm can be predicted by separate models. When both outputs are desired, however, a shared encoder with task-specific heads can reuse one representation rather than duplicate encoders. This convenience may couple the tasks. Machine-generated text detection (MGTD) must distinguish human-written from AI-generated text and thus favors source-predictive representations (Guo et al., 2023; Wang et al., 2024) (§2.2); these stylistic or distributional cues may overlap with features used for sarcasm detection. The labels do not conflict; the potential conflict lies in how the tasks organize a shared representation. MGTD must encode source, whereas sarcasm detection should reflect intent rather than source. Shared-encoder multi-task learning can suffer task conflict and negative transfer by exploiting information useful for one task but spurious for another (Mueller et al., 2022; Hu et al., 2022b). We therefore test whether joint training changes sarcasm predictions.

Source-predictive regularities and sarcasm cues may vary across language and corpus settings (Jang and Frassinelli, 2024; Casola et al., 2024; Wang et al., 2024). We therefore conduct the diag-

nosis in Chinese and English and report language-stratified results to assess the robustness of the subgroup pattern across the two settings.

To test how joint MGTD training changes sarcasm behavior, we compare same-architecture dual- and single-head models and build DualTest for source $\times$ sarcasm subgroup analysis. The evaluated joint-training procedure sharply increases the requested-style false-positive rate on AI text prompted to be non-sarcastic; we call this class-conditional negative transfer. Within AI-generated texts, relative ranking remains largely preserved; joint training instead shifts sarcasm scores upward.

We then examine the shared representation. Source information is highly decodable and correlates with sarcasm decisions. To connect encoded information to actual decisions beyond probe accuracy (Hewitt and Liang, 2019; Elazar et al., 2021), we combine probes, task-head correlations, and representation interventions. These analyses identify a non-random high-variance direction to which the subgroup gap is sensitive, supporting a source-correlated shortcut hypothesis and providing a focused target for further causal analysis (§5.3.2, Appendix E).

Under fixed-detector evaluation of Qwen3.5-generated replacements, the matched plain dual-single contrast retains the same direction (§5.3.3).

The dual-head design also has a downstream implication. Detector scores can become generation rewards: ViSP derives PPO rewards from a sarcasm detector (Wang et al., 2026). Yet useful generated sarcasm must satisfy two distinct objectives: conveying sarcastic intent and being perceived as human-like. A dual-head detector may appear to provide corresponding signals through sarcasm and predicted human-source probabilities, but predicted human-source probability is not itself a measure of human-likeness. We examine this implication exploratorily by comparing detector scores with source-label-blinded LLM ratings of sarcasm and perceived human-likeness (§6).

Contributions. We construct DualTest from Chinese and English resources for source $\times$ sarcasm diagnosis of joint MGTD-sarcasm training. Same-architecture dual-single comparisons reveal class-conditional negative transfer under the evaluated joint-training procedure, concentrated on AI texts prompted to be non-sarcastic and characterized by a source-conditioned score shift; the direction persists under fixed-detector evaluation of a

second generator. Representation diagnostics support a source-correlated shortcut hypothesis and identify a focused target for further causal analysis. An exploratory detector-judge comparison illustrates the risk of treating detector probabilities as measures of sarcasm quality or human-likeness.

2 Related Work

2.1 Sarcasm Detection, Synthetic Data, and Multilingual Resources

Feature-engineered methods (Riloff et al., 2013), neural sequence models (Ghosh and Veale, 2016; Tay et al., 2018), and pretrained language models (Khatri and P, 2020; Du et al., 2022) have each advanced sarcasm detection. Performance nevertheless varies substantially across datasets because label definitions, domains, languages, and modalities differ, particularly in whether labels reflect intended or annotator-perceived sarcasm (Oprea and Magdy, 2020; Abu Farha et al., 2022; Casola et al., 2024; Yue et al., 2024).

Recent work evaluates LLM-generated sarcasm directly: Majer et al. (2026) use predictive, linguistic, and human-judgment criteria, while Mahato et al. (2026) link synthetic-data signatures to shortcut learning and poor transfer to human-authored texts. Building on these findings, we study shared-encoder joint MGTD-sarcasm training against same-architecture sarcasm-only controls.

RCNNs combine recurrent context with convolutional pooling for text classification (Lai et al., 2015) and have also been evaluated for multilingual sarcasm detection (Yang and Ikeda, 2024); we therefore include an RCNN intermediate configuration in §4.1.

DualTest operationalizes this question with topic-matched human/AI data spanning all four source $\times$ sarcasm groups in Chinese and English; §5.1 reports directly comparable mBERT dual-head baselines under a shared evaluation protocol.

2.2 Machine-Generated Text Detection and Authorship Verification

MGTD comprises zero-shot model-likelihood methods, such as probability-curvature detectors (Mitchell et al., 2023), and discriminative models trained on paired human-AI corpora (HC3; Guo et al., 2023). Generalization is a central difficulty: M4 (Wang et al., 2024) provides a benchmark across generators, domains, and languages and

shows that detectors struggle on unseen domains or LLMs, while RAID (Dugan et al., 2024) stress-tests detectors under unseen generators, decoding variations, and adversarial attacks. This generalization gap is consistent with reliance on source-predictive regularities that are partly generator- or domain-specific; such cues are relevant here because they may enter a representation shared with sarcasm detection. Bevendorff et al. (2025) distinguish closed-set authorship attribution from open-set authorship verification and argue that realistic LLM detection is better cast as the latter. Our binary MGTD setup is narrower: we use it as a controlled source-classification auxiliary task, not as a measure of text quality.

We are not aware of prior work that directly studies shared-encoder joint training for sarcasm and MGTD.¹ This gap matters because source information has different roles in the two tasks: it is the target of MGTD but should not by itself determine a sarcasm detector’s prediction. Our study tests whether a shared representation preserves that distinction.

2.3 Multi-Task Negative Transfer, Shortcut Learning, and Shared-Encoder Mitigation

In multi-task learning, auxiliary tasks are not always beneficial: transfer depends on task relatedness and alignment between their data distributions (Zhang et al., 2023; Wu et al., 2020), while gradient conflict is a central optimization challenge that can contribute to negative transfer (Yu et al., 2020; Liu et al., 2021). Mueller et al. (2022) provide a relevant NLP precedent through matched single- and multi-task comparisons. They find similar negative-transfer patterns in canonical and text-to-text systems and note that models may infer task identity from input distributions, a strategy that may not transfer reliably out of distribution. He and Choi (2021) and Li et al. (2023) further analyze functional specialization in multi-task Transformer encoders. Shortcut learning offers a complementary reading: models may exploit features that are predictive in the training distribution yet fail under distribution shift (Geirhos et al., 2020; Ye et al., 2026). In MTL, shared representations

can increase reliance on spurious correlations (Hu et al., 2022b; Chai et al., 2025). Related multi-dataset work shows that adding data can introduce spurious correlations and reduce worst-group performance (Compton et al., 2023).

Shared-encoder mitigation either isolates task-specific parameters (Pfeiffer et al., 2021, 2020; Karimi Mahabadi et al., 2021; Stickland and Murray, 2019) or modifies information flow within a shared computation graph (Wu et al., 2019; Pilault et al., 2021). Our analysis is closer to the second design direction: we compare a no-module baseline with two intermediate-representation variants and test whether degradation changes across source $\times$ sarcasm subgroups (§5.3).

2.4 Misalignment in Detector-Based Evaluation and Rewards

The representational conflict studied above may also matter when detector outputs are reused beyond classification, particularly as evaluation or reward signals for generated text. Automated scorers, including LLM judges, are increasingly used to evaluate generated text (Zheng et al., 2023). Model-based scores are also optimized as rewards, but overoptimization can increase a proxy score while degrading a gold-standard measure (Gao et al., 2023). A detector can achieve useful classification accuracy without its continuous output being calibrated as a measure of style strength or generation quality. Multimodal sarcasm generation may face this risk: ViSP (Wang et al., 2026) uses the DIP sarcasm detector (Wen et al., 2023) as a PPO reward (Schulman et al., 2017). As an exploratory downstream analysis, Section 6 compares predicted human-source probabilities with separately obtained, source-label-blinded LLM ratings of perceived human-likeness.

3 Dataset Construction

Testing for interference requires separate task supervision and evaluation labels for source and sarcasm. Here “source” denotes whether a text is human-authored or AI-generated; “corpus” denotes an input dataset. We combine iSarcasm (Oprea and Magdy, 2020; Abu Farha et al., 2022), CSC (Jang and Frassinelli, 2024; Jang et al., 2023), ToSarcasm (Liang et al., 2022), News Headlines v2 (Misra and Arora, 2023; Misra and Grover, 2021), and the Open Chinese Internet Sarcasm Corpus (Zhu, 2022). None of these datasets provides

¹Searches conducted in May 2026 and updated in July 2026 across Semantic Scholar, arXiv, ACL Anthology, and Google Scholar, using keyword combinations including “sarcasm detection,” “synthetic sarcasm,” “LLM-generated sarcasm,” “machine-generated text detection,” “AI-generated text detection,” “multi-task learning,” and “joint detection.”

both labels. Mapping their annotations to a common binary sarcasm label yields 44,862 pooled instances (approximately 87% English). From this pool, we construct separate resources for sarcasm supervision (SarcTrain), source supervision (AuthTrain), generation fine-tuning (GenTrain), and source $\times$ sarcasm evaluation (DualTest; Appendix A).

SarcTrain. SarcTrain contains 10,000 human-authored texts for sarcasm classification. Each language $\times$ label bucket contains 2,500 instances. Three buckets are downsampled; the Chinese sarcastic bucket starts from 2,396 records and adds 104 oversampled records. The resulting set is organized into five folds.

AuthTrain. AuthTrain contains 19,995 texts for source classification. After seed-level generation quality control excludes one Chinese SarcTrain instance from use as a seed, AuthTrain retains 9,999 approximately language-balanced human seeds (5,000 English / 4,999 Chinese). The Olama model tagged deepseek-r1:8b (DeepSeek-AI, 2025; Ollama, 2025) produces 9,996 retained AI counterparts; three selected human seeds remain unmatched, giving 99.97% pairing coverage. Appendix A details the exclusions.

We use four language $\times$ style templates, with each seed’s label determining the requested style. Source and requested-style labels form two separately controlled, approximately balanced factors, enabling source $\times$ sarcasm subgroup analysis (Appendix A).

GenTrain. GenTrain contains 2,428 texts for the supervised fine-tuning (SFT) used in §6. From 5,257 eligible positive-sarcasm records in iSarcasm and ToSarcasm, we sample 2,103 records (40%; 1,214 English and 889 Chinese). Chinese oversampling adds 325 records, yielding 1,214 per language. The resulting set contains 374 rewriting, 839 response, and 1,215 comment instances (Appendix A).

DualTest. DualTest contains 4,664 texts for source $\times$ sarcasm evaluation. Its human subset combines official iSarcasm (1,400 English) and ToSarcasm (972 Chinese) test sets. After generation quality control, AI counterparts remain for 2,292 of 2,372 texts (96.6% coverage). Human texts retain their annotations; for AI texts, the sarcasm label records the prompted condition rather

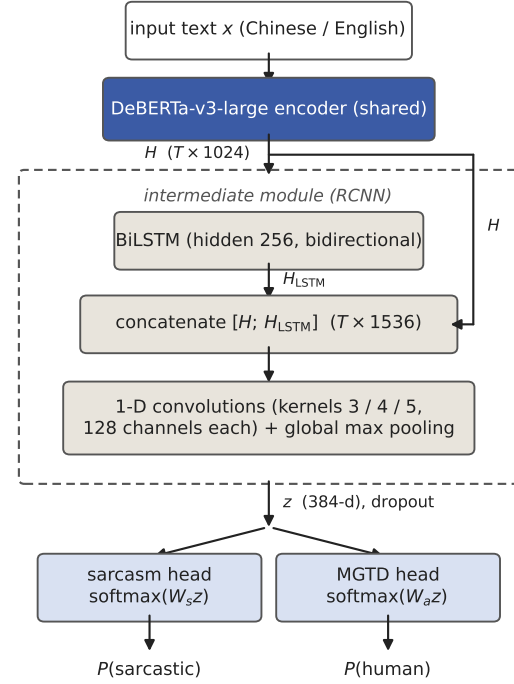


Figure 1: RCNN dual-head configuration: a shared DeBERTa-v3-large encoder, an RCNN intermediate module (dashed), and two task-specific heads. The comparisons in §5 remove this component in the plain configuration or replace it with a task-branch interaction-projection module, while retaining the same encoder backbone and task definitions (Appendix B).

than an independently annotated judgment of perceived sarcasm.

The four source $\times$ sarcasm groups contain 823 human-sarcastic, 1,549 human-non-sarcastic, 779 AI-sarcastic, and 1,513 AI-non-sarcastic instances. Appendix A gives the language $\times$ source $\times$ sarcasm distribution.

4 Joint and Single-Task Models

4.1 Tasks and Architecture

Given a Chinese or English input text x , the model produces two separate, non-mutually-exclusive binary predictions: whether x is labeled sarcastic and whether its source label is human or AI. Any combination of the two labels is valid. The tasks share a single DeBERTa-v3-large encoder (He et al., 2023), allowing us to test how incorporating MGTD into joint training changes the representation read by the sarcasm head. We compare three shared-representation configurations and examine how subgroup errors vary across them; Figure 1 shows the RCNN configuration.

The RCNN configuration uses BiLSTM + CNN as its intermediate module. A single-layer bidirectional LSTM with hidden size 256 per direction produces a 512-dimensional output at each token position. We concatenate this output with the 1,024-dimensional encoder output, then apply three groups of one-dimensional convolutions with kernel widths 3/4/5 and 128 channels each. Global max pooling produces a 384-dimensional vector z . After dropout, z is passed to the two heads, yielding a DeBERTa-based dual-head RCNN configuration.

Alongside RCNN, §5 compares two further configurations. The plain model sends [CLS] directly to both heads. The interaction-projection model forms two learned 256-dimensional projections, h_s and h_a , and updates them as $h'_s = h_s + P_{a \rightarrow s}(h_a)$ and $h'_a = h_a + P_{s \rightarrow a}(h_s)$. Because each branch supplies only one key-value vector, every attention head assigns a weight of one to that vector; the module therefore reduces to residual cross-branch affine projections, where $P_{a \rightarrow s}$ and $P_{s \rightarrow a}$ each compose the attention value and output projections. The configurations retain the same encoder and task definitions while testing distinct intermediate designs (Appendix B).

4.2 Training Protocol

The training protocol supports controlled dual-single comparisons in the plain and RCNN configurations. Within each comparison, the models use the same encoder architecture, intermediate configuration, sarcasm data and fold assignments, and sarcasm-specific hyperparameters. The dual model additionally receives AuthTrain batches, updates shared parameters from both tasks, follows the mixed-task schedule, and uses joint early stopping; the comparison therefore evaluates the complete joint-training procedure rather than any single component.

Training uses a mixed loader that supplies task-homogeneous batches according to a 2:1 MGTD:sarcasm schedule, matching the approximate data-size ratio between AuthTrain and SarcTrain. Let θ_{sh} denote parameters updated by both tasks, let $z_s(x)$ and $z_a(x)$ be the representations read by the sarcasm and MGTD heads, and let W_s and W_a denote the two classification heads. The two representations coincide in the plain and RCNN configurations; in the interaction-projection configuration, they are the task-specific h'_s and h'_a defined above. At step t , the loss is com-

puted only for the task corresponding to the current batch B_t :

$$\mathcal{L}(B_t) = \begin{cases} \frac{1}{|B_t|} \sum_{(x,y) \in B_t} \text{CE}_w(W_s z_s(x), y), & B_t \subset \text{SarcTrain}, \\ \frac{1}{|B_t|} \sum_{(x,y) \in B_t} \text{CE}(W_a z_a(x), y), & B_t \subset \text{AuthTrain}, \end{cases} \quad (1)$$

where CE_w is a pre-specified cost-sensitive cross-entropy for sarcasm (weights are provided in Appendix B). For the class-balanced SarcTrain dataset, the weighted loss assigns a greater penalty to errors on sarcastic examples. Softmax is applied to logits only when probabilities are required. Each task-homogeneous batch updates the active task head and the shared parameters. Unless otherwise stated, binary predictions use a fixed probability threshold of 0.5 rather than a threshold tuned on DualTest.

The encoder uses a lower learning rate than newly added modules. The sarcasm-head rate is fixed across configurations; MGTD-head rates are reported in Appendix B.

Model selection accounts for both tasks. At epoch e , the early-stopping signal is the equally weighted sum of the two validation losses, each normalized by its first-epoch value:

$$\mathcal{L}_{\text{stop}}^{(e)} = \frac{1}{2} \frac{\mathcal{L}_s^{(e)}}{\mathcal{L}_s^{(1)}} + \frac{1}{2} \frac{\mathcal{L}_a^{(e)}}{\mathcal{L}_a^{(1)}}, \quad (2)$$

where $\mathcal{L}_s^{(e)}$ and $\mathcal{L}_a^{(e)}$ are the epoch- e validation losses for the sarcasm and MGTD tasks on their respective validation sets. Normalization reduces the influence of differences in task-specific loss scales and learning dynamics.

Five runs are defined by the SarcTrain validation folds: four folds are used for sarcasm training and one for validation, while AuthTrain uses a fixed 90/10 split. Each run reloads the pre-trained encoder and randomly initializes new modules and heads. Configurations share fold assignments and nominal seeds, enabling split-matched comparisons but not identical initialization.

DualTest is not included in any training or validation split for either task. Five-run standard deviations combine fold and initialization variation. Wilson intervals are conditional on a fixed final model and cover test-instance, not training or generator, uncertainty (Table 3; Appendix C.7).

For each configuration used in the subgroup analysis, a final model is trained on all applicable data with a fixed epoch count determined from the

five validation runs, an independent seed, and no early stopping. DualTest is not used to train these models or determine their training duration, and the final models are not included in five-run aggregates; §6 uses the final RCNN dual-head model (Appendix B).

5 Experiments and Diagnosis

With the sarcasm and MGTD models fixed, we use DualTest first to compare encoder backbones and intermediate configurations, then to measure the change between the joint- and single-task procedures, and finally to partition predictions into source $\times$ sarcasm groups. Aggregate results are five-run means $\pm$ standard deviations; fixed final models support paired instance-level subgroup comparisons and Wilson intervals.

5.1 Backbone Comparison

We compare dual-head mBERT (Devlin et al., 2019), XLM-RoBERTa-base (Conneau et al., 2020), DeBERTa-v3-base, and DeBERTa-v3-large (He et al., 2023) under one evaluation protocol, adding the two intermediate variants on the large encoder. Positive-class F1 is computed for the sarcastic class in sarcasm detection and the human class in MGTD over the full DualTest set.

Table 1 shows higher mean F1 for the DeBERTa configurations than for mBERT and XLM-R. Among the evaluated configurations, RCNN obtains the highest means, reaching 0.6309 sarcasm F1 and 0.9301 MGTD F1. The two intermediate variants also show smaller standard deviations across the five runs than the plain large model. Because backbone identity, scale, and intermediate representation differ jointly, these results support only configuration-level comparisons. For the separately trained final RCNN model, sarcasm F1 is 0.819 in Chinese and 0.429 in English; prevalence, genre, annotation, and language may all contribute to this difference, so it should not be read as a pure language effect. Appendix C provides macro-averaged, human-subset, and language-stratified results.

5.2 Dual-Head versus Single-Head Comparison

Within each architecture, dual- and single-head models share the encoder, intermediate structure, sarcasm data, fold assignments, and sarcasm-specific hyperparameters; dual runs additionally receive AuthTrain batches and MGTD updates.

Model	Sarc. F1	MGTD F1
mBERT dual-head	0.4945 $\pm$.036	0.8743 $\pm$.006
XLM-RoBERTa-base dual-head	0.4603 $\pm$.057	0.8321 $\pm$.025
DeBERTa-v3-base dual-head	0.5646 $\pm$.056	0.8952 $\pm$.004
DeBERTa-v3-large plain dual-head	0.6126 $\pm$.031	0.9221 $\pm$.010
+ interaction projection	0.6076 $\pm$.020	0.9126 $\pm$.007
+ RCNN	0.6309 $\pm$.020	0.9301 $\pm$.005

Table 1: Descriptive dual-head results across backbones and architectural variants (DualTest; five-run mean $\pm$ standard deviation; bold indicates the highest mean).

Arch.	Task	Dual	Single	Δ
Plain	Sarc.	0.6126 $\pm$.031	0.6734 $\pm$.010	-6.08
Plain	MGTD	0.9221 $\pm$.010	0.9054 $\pm$.027	+1.67
RCNN	Sarc.	0.6309 $\pm$.020	0.6751 $\pm$.022	-4.43
RCNN	MGTD	0.9301 $\pm$.005	0.9253 $\pm$.006	+0.49

Table 2: Dual-head vs. single-head models (mean $\pm$ standard deviation over five fold-based runs; Δ = dual – single, in percentage points).

Single-head controls process the full sarcasm training fold each epoch, whereas dual-head models draw sarcasm batches under the 2:1 schedule (§4.2).

We use negative transfer to denote degradation relative to a same-architecture single-task control. Joint training lowers sarcasm F1 in all ten split-matched comparisons—five per architecture—with differences from -0.8 to -8.6 pp. Mean sarcasm F1 decreases by 6.08 pp for the plain model and 4.43 pp for RCNN, whereas mean MGTD F1 changes by only +1.67 and +0.49 pp. Only sarcasm detection shows consistent negative transfer across all ten split-matched comparisons. We next identify which source and sarcasm groups account for it. RCNN shows a smaller dual–single gap.

5.3 Source $\times$ Sarcasm Subgroup Analysis

The aggregate F1 decrease is highly uneven across source $\times$ sarcasm groups. We therefore examine the four DualTest quadrants separately. Human texts retain dataset sarcasm annotations; for AI texts, the sarcasm labels denote requested styles. Because each group contains only one reference class, we report recall for sarcastic texts and false-positive rate for non-sarcastic texts.

5.3.1 Three Architectures and Single-Head Controls

The three dual-head variants characterize how subgroup errors vary across intermediate representations. Plain and RCNN single-head models provide same-architecture controls. Within each pair, both models use the same all-human sarcasm data; only the dual-head model is additionally trained on MGTD. Their difference therefore measures the complete joint-training procedure. Table 3 reports sarcasm predictions; corresponding MGTD results are in Appendix C.

False-positive rates on AI $\times$ non-sarcastic instances are 86.3% for the plain dual-head model, 40.9% for interaction projection, 34.1% for RCNN, and 19.3% for both same-architecture single-head controls; the intermediate rates are therefore more than 45 pp below the plain rate.

On the same AI non-sarcastic texts, 1,021 false positives occur only under the plain dual-head model, versus 7 only under its single-head control ($p < .001$). The control receives the same all-human sarcasm supervision but is not jointly trained on MGTD, tying the large increase to the joint-training procedure under comparison. RCNN shows the same direction with a smaller gap. The plain and RCNN models provide the matched contrasts, while interaction projection provides a complementary comparison across configurations.

No analogous extreme subgroup error appears in the MGTD head: its false-positive rate is 6.1% for plain and 6.6% for RCNN. Within AI-generated texts, sarcasm AUC remains 0.82–0.88 across models, so the relative ranking of sarcastic and non-sarcastic AI texts is largely preserved. Joint training instead shifts predicted sarcasm probabilities upward, causing many texts prompted to be non-sarcastic to cross the fixed 0.5 threshold. This source-conditioned score shift, rather than a collapse of within-AI ranking ability, characterizes the subgroup failure.

The matched plain comparison makes the concentration especially clear: its non-sarcastic false-positive rate changes by 4.2 pp on human text but by 67.0 pp on AI-generated text. The aggregate F1 decrease is therefore driven by a sharply localized source $\times$ class effect rather than a uniform loss across DualTest.

5.3.2 Representation-Level Diagnostics

We next ask whether source information in the shared representation is associated with the higher sarcasm scores. A linear probe predicts source from both dual-head (AUC 0.986) and same-architecture single-head (0.974) representations. Source decodability alone therefore does not explain the dual–single gap. In the dual-head model, texts with a higher MGTD-head AI logit also tend to have a higher sarcasm logit in the target subgroup ($\rho = 0.519$). After covariance normalization, the source directions estimated from the MGTD head and the probe are also moderately aligned with the sarcasm direction. Together, the logit correlation and directional alignment provide correlational evidence linking source-related structure to the affected sarcasm decisions.

We then modify the frozen representations and recompute predictions. LEACE (Belrose et al., 2023) substantially reduces linear source decodability, lowering source-probe AUC from 0.982 to 0.589, but mainly reallocates errors between source groups. The AI/human non-sarcastic false-positive rates change from 84.8%/32.8% to 68.7%/49.5%, while their unweighted mean remains nearly unchanged (58.8% to 59.1%).

A key control removes a high-variance direction constructed to be orthogonal to the estimated linear source direction. This changes the AI/human false-positive rates to 66.7%/54.1% even though measured source decodability remains 0.982; ten sampled random directions show no comparable change. Together, these controls identify a non-random high-variance direction to which source-group errors are sensitive. Combined with the preceding logit correlations and directional alignment, the results support a source-correlated shortcut hypothesis and provide a focused target for further causal analysis (Appendix E).

5.3.3 Fixed-Detector Evaluation on a Second Generator

We keep the trained detectors fixed and compare the DualTest AI texts generated by deepseek-r1:8b with replacements generated by Qwen3.5-9B (Ollama tag qwen3.5:9b) (Qwen Team, 2026). After quality control, the non-sarcastic subsets generated by deepseek-r1:8b and Qwen3.5-9B contain 1,513 and 1,550 outputs. The plain dual-head false-positive rates are 86.3% and 91.3%, whereas the corresponding single-head rates are 19.3% and 7.9%; the dual–single

Model	H sarc. rec.	H non-sarc. FP	AI sarc. rec.	AI non-sarc. FP
Plain dual-head	67.1%	32.9%	98.2%	86.3% [84.5, 88.0]
Interaction-projection dual-head	71.9%	37.6%	94.4%	40.9% [38.5, 43.4]
RCNN dual-head	80.8%	40.5%	89.3%	34.1% [31.8, 36.5]
Plain single-head (sarcasm only)	67.8%	28.7%	76.6%	19.3% [17.4, 21.4]
RCNN single-head (sarcasm only)	66.2%	28.5%	79.1%	19.3% [17.4, 21.4]

Table 3: Sarcasm predictions across four source $\times$ sarcasm groups. Human texts use dataset sarcasm annotations; AI-side sarcasm labels denote requested styles. Bold marks the focal AI non-sarcastic subgroup. Brackets show Wilson 95% intervals for the fixed final models; full intervals and paired tests are in [Appendix C.7](#).

gap therefore changes from 67.0 to 83.4 pp. On the Qwen3.5-generated subset, the RCNN and interaction-projection rates remain much lower than the plain dual-head rate, at 22.1% and 29.5%. The dual-single contrast retains the same direction on Qwen3.5 outputs, and the three dual-head configurations retain the same ordering, although the magnitudes differ (Appendix C).

6 Exploratory Comparison of Detector Scores and LLM-Judge Ratings

As a downstream test, we compare the RCNN detector’s human-source and sarcasm probabilities with source-label-blinded LLM ratings of human-likeness and sarcasm. We evaluate 600 outputs from Qwen3-8B (Yang et al., 2025) before and after QLoRA (Dettmers et al., 2023) fine-tuning on GenTrain. DeepSeek-V3 (DeepSeek-AI, 2024), Gemini-2.5-Flash-Lite (Google DeepMind, 2025), and GPT-5-nano (OpenAI, 2025) rate outputs from 180 Chinese topic-comment and 120 English rewriting prompts; language is therefore coupled with generation task. Because predicted human-source probability and perceived human-likeness are distinct constructs, this comparison tests whether classifier scores can serve as generation-evaluation proxies.

Detector probabilities and judge ratings do not move consistently. In Chinese, SFT raises the RCNN detector’s sarcasm score from 0.780 to 0.895 but lowers the mean LLM sarcasm rating from 3.99 to 3.80; the detector–judge correlation is -0.020 in Chinese and $+0.338$ in English. SFT also raises predicted human-source probabilities in both task–language settings while mean perceived human-likeness ratings decline, yielding a pooled correlation of -0.212 ; weaker within-condition correlations show that this value is driven mainly by the condition-level reversal.

The sarcasm-detector change is setting-dependent ($+0.115$ in Chinese; -0.124 in

English), whereas SFT receives lower mean sarcasm ratings in all six language $\times$ judge comparisons, five of which remain significant after FDR correction. Detector probabilities can therefore remain useful classification outputs without preserving the ordering or condition-level direction required of style or quality scores.

Individual-text judge agreement is low to moderate; a plain-detector control changes correlations by at most 0.02. The human-source/human-likeness reversal and judge-specific sarcasm results motivate external validation before detector probabilities are reused as quality scores or rewards; [Appendices C–D](#) report full protocols, tests, and cases.

7 Conclusion

Our results identify a source-conditioned negative-transfer pattern in joint MGTD–sarcasm training: relative to a same-architecture single-head control, the plain dual-head model raises the requested-style false-positive rate on AI text prompted to be non-sarcastic by 67.0 pp. Within AI-generated texts, relative ranking remains largely preserved; the subgroup effect is an upward shift in sarcasm scores rather than a collapse of within-source discrimination. The plain dual-single contrast has the same direction in both languages and under fixed-detector evaluation of Qwen3.5-9B outputs, while RCNN and interaction projection yield lower subgroup rates in the evaluated configurations. Representation diagnostics support a source-correlated shortcut hypothesis and identify a focused target for further causal analysis. Predicted human-source probabilities and human-likeness ratings diverge, motivating external validation before detector scores are used as quality metrics or rewards.

Limitations

DualTest combines human sarcasm annotations with requested-style sarcasm labels for AI texts,

so AI-side false positives are defined relative to requested style rather than independent perceived-sarcasm annotation.

The core dual–single comparisons evaluate the complete joint-training procedure: architecture, sarcasm data, folds, sarcasm-head learning rate, and class weights are shared, whereas MGTD batches, task scheduling, shared updates, and joint early stopping are introduced together. Compute-matched controls could further decompose this effect. Cross-architecture results are configuration-level because capacity and optimization differ, and the cost-sensitive loss can affect absolute rates.

Because oversampling precedes fold assignment, some SarcTrain copies occur in different folds and can appear on both sides of a training–validation split, introducing dependence into checkpoint selection. The affected records are absent from DualTest. Generation failures from `deepseek-r1:8b` leave 80 human texts without generated counterparts. AuthTrain is generated with `deepseek-r1:8b`; Qwen3.5-9B replaces only DualTest’s AI side and therefore provides test-side directional evidence rather than a retraining-based replication.

The fixed-model linear analyses connect source-related structure to behavior but do not isolate a unique causal feature or characterize variation across training seeds. The detector–judge study uses ratings from LLMs rather than humans and couples language with generation task. Future work should test these findings under independent style annotation, matched training budgets, and broader human and cross-generator evaluation.

Ethical considerations

All human-authored texts come from publicly released research datasets: iSarcasm, CSC, ToSarcasm, News Headlines, and the Open Chinese Internet Sarcasm Corpus. Explicit repository licenses were available for iSarcasm and CSC; for sources without an identified redistribution license, we do not redistribute the human-authored text (Appendix A). We collect no new user data. We did not conduct any additional audit of the released corpora for personally identifiable information or offensive content.

All AI texts were generated locally using publicly released open-weight models (`deepseek-r1:8b`, Qwen3.5-9B, and the base and QLoRA-adapted Qwen3-8B). They are used

only for training and evaluation and are labeled as AI-generated. No new human participants were recruited; the study analyzes human-authored texts from previously released datasets. The automated evaluation transmitted each evaluation input and generated output, together with rating instructions, to the DeepSeek-V3, Gemini-2.5-Flash-Lite, and GPT-5-nano APIs; it did not transmit dataset metadata or ground-truth source labels.

The detectors were developed for research diagnosis. Their outputs should neither be used as evidence of authorship or misconduct at the individual-text level without human review and domain-specific validation nor be deployed for content moderation or automated sanctions without evaluating out-of-distribution behavior. The class-conditional failure diagnosed here is an example of systematic subgroup error affecting a particular type of generated content.

AI-assisted tools were used to support language editing, structural revision, and formatting. They did not generate the reported experimental outputs or statistics; the authors determined the study design, analysis, and conclusions and take full responsibility for the work.

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A Detailed Dataset Statistics and Quality Checks

A.1 Corpus details. The pooled collection contains 44,862 instances in total (86.9% English / 13.1% Chinese). Chinese–English 1:1 balance in SarcTrain is achieved through bucketed sampling. The two English buckets and the Chinese non-sarcastic bucket contain more than 2,500 available instances and are randomly downsampled within each bucket. The Chinese sarcastic

Corpus	Lang.	Inst.	Use
iSarcasm (SemEval-2022 T6)	English	3,467	Sarc. train + SFT
CSC	English	6,878	Sarc. train
ToSarcasm	Chinese	3,898	Sarc. train + SFT
News Headlines v2	English	28,619	Sarc. train
Open Chinese Internet Sarcasm Corpus	Chinese	2,000	Sarc. train

Table 4: Input corpora. Genre / annotation: iSarcasm — tweets, speaker self-annotation; CSC — conversational responses; ToSarcasm — topic comments, third-party annotation; News Headlines — distant supervision (The Onion / HuffPost); Open Chinese Corpus — Weibo, third-party annotation.

bucket contains 2,396 source records representing 2,394 unique texts; 104 records are added by oversampling to reach 2,500. When source-level text duplicates are counted, the final bucket’s text-duplication rate is 4.24% (A.3).

Licensing and release scope. The iSarcasm and CSC repositories specify the MIT License. We did not identify explicit redistribution licenses in the releases we accessed for ToSarcasm, News Headlines v2, or the Open Chinese Internet Sarcasm Corpus. We therefore do not redistribute human-authored text from those sources. Planned research artifacts are limited to eligible AI-generated text, labels and metadata, and reconstruction scripts that require users to obtain the human-authored corpora from their original sources.

Lang.	Source	Sarc.	Non-sarc.
English	Human	200	1,200
English	AI	192	1,174
Chinese	Human	623	349
Chinese	AI	587	339

Table 5: Language $\times$ source $\times$ sarcasm distribution of DualTest. Human texts use dataset sarcasm annotations; AI-side sarcasm labels denote requested styles.

A.2 Three-dimensional distribution of DualTest.

A.3 Data quality checks. Automated checks confirm language and label balance across the five SarcTrain folds. AuthTrain contains 9,996 complete human–AI pairs and three unmatched human instances. Record-level matching confirms that

neither SarcTrain nor AuthTrain shares records with the human or AI side of DualTest; no train–test leakage is present.

SarcTrain’s text-duplication rate is 0.34% for English / 2.16% for Chinese. Chinese duplication is concentrated in the sarcastic bucket (4.24%; non-sarcastic bucket: 0.08%) and results from the oversampling used to fill that bucket (A.1). Because oversampling precedes fold assignment, 86 of the 104 added records have a source counterpart in another fold. Under the raw-text counting convention used for these duplication rates, 116 repeated-text groups involving 241 records are present when oversampled copies are included, and 98 groups span folds. Excluding the added copies and normalizing whitespace and case leaves 19 naturally duplicated groups involving 46 records, of which 16 span folds. These relationships are relevant to fold-based checkpoint selection but do not place the affected training or validation records into DualTest. Post-processing also checks for the explicit chain-of-thought markers and reasoning prefaces described in A.4 (Wei et al., 2022).

A.4 AI generation protocol. The AI texts in AuthTrain and the two DualTest versions are produced by locally deployed generation models through the Ollama API. The model tagged deepseek-r1:8b generates AuthTrain and the DeepSeek version of DualTest’s AI side; the Qwen3.5-9B model tagged qwen3.5:9b generates the alternate AI side used for fixed-detector evaluation on a second generator. Both models use temperature = 0.8, top- p = 0.95 (Holtzman et al., 2020), and a generation limit of 500 tokens. Generation instructions use four templates defined by language $\times$ style, with style determined by the seed instance’s sarcasm label. The templates constrain output style and length (20–80 Chinese characters; 10–50 English words). These constraints are intended to reduce, but do not eliminate, source differences in length; Appendix E separately repeats the representation correlation analysis within length tertiles. The complete Chinese sarcastic template used in the experiment is reproduced verbatim below:

【任务】生成一条讽刺风格的评论
 【主题】{seed}
 【要求】
 - 直接给出简短的评论，不要解释思路或分析
 - 风格要讽刺、幽默、夸张
 - 表达与主题相关的观点即可，不必追求严谨
 - 20–80 字为佳
 【评论】

Its pinyin transliteration is:

[rènwù] shēngchéng yī tiáo fēngcì
fēnggé de pínglùn
[zhǔtí] {seed}
[yāoqiú]
- zhíjiē gěichū jiǎnduǎn de pínglùn,
bùyào jiěshì sīlù huò fēnxī
- fēnggé yào fēngcì, yōumò, kuāzhāng
- biǎodá yǔ zhǔtí xiāngguān de guāndiǎn
jíkě, bùbù zhuīqiú yánjīn
- 20–80 zì wéi jiā
[pínglùn]

Its English translation is:

[Task] Generate a sarcastic-style
comment
[Topic] {seed}
[Requirements]
- Directly provide a short comment;
do not explain your reasoning or
analysis
- The style should be sarcastic,
humorous, and exaggerated
- It is sufficient to express a
viewpoint related to the topic;
strict rigor is not required
- Preferably 20--80 Chinese
characters
[Comment]

The Chinese non-sarcastic template changes only the style line to “The style should be normal and natural.” The English templates follow the same structure, using either “sarcastic, humorous, exaggerated” or “normal, natural,” with a length limit of 10–50 words.

Post-processing removes `<think>...</think>` segments, reasoning prefaces such as “好的” (*hǎo de* ‘okay’), “嗯” (*ń* ‘hmm’), “Okay,” “Let me,” and “First,” and Markdown markup. Outputs that still exceed the length limit or exhibit cyclic repetition after repeated generation attempts are discarded. AuthTrain retains 9,996 AI counterparts from 9,999 seeds; DualTest retains 2,292 counterparts from 2,372 seeds.

The Qwen3.5-9B replacement set has no generation failures, yielding 1,550 AI $\times$ non-sarcastic instances rather than 1,513. Generator-specific rates therefore use the full set of retained outputs for each generator rather than an identical seed intersection. For this run, Ollama thinking mode is disabled (`think = false`) to prevent the reasoning process from exhausting the output budget. The same post-processing removes any inline chain-of-thought from the `deepseek-r1:8b` outputs.

A.5 SFT training and generation-evaluation set.

Eligibility filtering retains 5,257 positive-sarcasm records suitable for generation tasks (2,972 English; 2,285 Chinese). After shuffling with seed 42, we sample 40% of these records, yielding 2,103 instances (1,214 English; 889 Chinese); language balancing then adds 325 oversampled Chinese records, producing GenTrain (2,428; 1,214 per language). By prompt type, it comprises 374 rewriting, 839 response, and 1,215 comment instances. Generation evaluation uses an independent set of 300 prompts (180 Chinese and 120 English): the Chinese task is to generate a sarcastic comment on a given topic, and the English task is to rewrite a normal sentence as a sarcastic one. This set is used for detector scoring and for all three source-label-blinded LLM evaluations in §6.

B Hyperparameters and Training Details

Group	Enc.	Inter.	Sarc.	MGTD
Backbone/plain dual	5×10^{-7}	—	5×10^{-6}	1×10^{-5}
RCNN dual	5×10^{-7}	5×10^{-6}	5×10^{-6}	5×10^{-6}
Inter.-proj. dual	5×10^{-7}	5×10^{-6}	5×10^{-6}	5×10^{-6}
Plain single-task	5×10^{-7}	—	5×10^{-6}	1×10^{-5}
RCNN single-task	5×10^{-7}	5×10^{-6}	5×10^{-6}	1×10^{-5}

Table 6: Discriminative learning rates for the dual- and single-task configurations.

B.1 Discriminative learning rates. The sarcasm-head learning rate is fixed at 5×10^{-6} across configurations. The MGTD-head rate is 1×10^{-5} for the plain dual and single-task models, 5×10^{-6} for the two intermediate dual-head variants, and 1×10^{-5} for the RCNN MGTD single-task control. Thus, the core sarcasm dual-single comparisons retain the same sarcasm-specific rates within each architecture, whereas the RCNN MGTD comparison and cross-configuration results can also reflect optimization settings. Early stopping uses the normalized joint validation loss of Equation 2 for dual-head models and the task’s own validation loss for the single-task controls, with the same patience of five epochs in both cases.

B.2 Shared training hyperparameters. Training uses AdamW with default PyTorch betas (0.9, 0.999) and $\epsilon = 10^{-8}$, weight decay 0.01, and parameter-group learning rates in B.1. A LambdaLR schedule linearly warms up for 200 steps

and then keeps the learning rate constant. Training runs for at most 10 epochs, with an early-stopping patience of 5 and gradient clipping at 1.0. The physical batch size is 4 with gradient accumulation over 4 steps, giving an effective batch size of 16. The MGTD:sarcasm sampling ratio is 2:1, and sarcasm class weights are [1.0, 2.5]. Both choices were specified before testing. For the class-balanced SarcTrain dataset, the class weights implement a cost-sensitive objective that assigns a greater penalty to errors on sarcastic examples. The normalized joint validation loss is $0.5 \times [\text{sarcasm validation loss} / \text{first-epoch value}] + 0.5 \times [\text{MGTD validation loss} / \text{first-epoch value}]$. The single-head controls process the full sarcasm training fold each epoch, whereas the dual-head models draw sarcasm batches under the 2:1 task-mixing schedule.

Detector inputs use a maximum sequence length of 512 tokens. In the RCNN module, a width- k convolution uses padding $\lfloor k/2 \rfloor$ followed by ReLU; global max pooling is applied directly over the full convolution output without an additional padding mask.

All fold-based runs use a seed equal to 42 plus the fold index and share the same fold assignments and nominal seed scheme across architectures and ablations. MGTD data use a fixed 90/10 holdout in every fold. For each final-model configuration, the fixed epoch count is the mode of its five best epoch counts plus one, capped at 10 epochs; the resulting counts range from 3 to 10 across configurations. The final model uses all applicable training data, a seed offset of +1000, no validation set, and no early stopping. Main tables report five-run means and standard deviations rather than final-model point estimates.

Model sizes, software, and hardware environment. The encoders have approximately 178M (mBERT), 278M (XLM-RoBERTa-base), 184M (DeBERTa-v3-base), and 435M (DeBERTa-v3-large) parameters; the SFT generator Qwen3-8B has about 8B parameters, of which QLoRA updates only low-rank adapters. Detector training and inference are performed on a single NVIDIA GeForce RTX 3090 (24 GB). Across the five-fold backbone, architecture, and ablation experiments and final-model training, total compute is on the order of tens of GPU-hours. A single detector run takes from a few minutes to a few hours, and SFT takes about two hours. The software environ-

ment is PyTorch 2.9.0 (CUDA 12.8), transformers 4.57.1, and peft 0.18.0. AI texts are generated locally through Ollama (Appendix A.4); blinded LLM evaluation uses API calls made in May 2026 (Appendix C.3).

B.3 SFT (QLoRA). We fine-tune Qwen3-8B using QLoRA (Dettmers et al., 2023), with 4-bit loading and bfloat16 computation. The LoRA configuration (Hu et al., 2022a) uses rank 64, alpha 16, and dropout 0 and targets q/k/v/o_proj plus gate/up/down_proj. The maximum sequence length is 2,048, with gradient checkpointing enabled. Training uses a learning rate of 5×10^{-5} , three epochs, batch size 1, gradient accumulation 8, and a warmup ratio of 0.05. After language balancing, a seed-42 split holds out 15% of GenTrain for validation; validation loss selects the checkpoint.

C Complete Experimental Results

C.1 Macro-averaged F1 (supplement to Table 1). Under the same five-run protocol, the RCNN macro-averaged F1 scores computed from each run’s confusion matrix are 0.6678 ± 0.036 for sarcasm and 0.9298 ± 0.005 for MGTD. Sarcasm macro-F1 is higher than its positive-class F1 (0.6309) because the majority class has a higher F1. These values supplement rather than rerank the full positive-class comparison in §5.1.

C.2 Complete four-quadrant results for the MGTD head (including single-head controls). The reporting convention matches Table 3 (recall for human subsets; false-positive rate for AI subsets).

Model	H×s	H×n	AI×s	AI×n
Plain dual	91.4	91.5	3.6	6.1
Inter.-proj. dual	85.4	85.4	1.8	4.1
RCNN dual	94.8	93.2	5.5	6.6
Plain single	90.4	86.6	4.1	4.2
RCNN single	90.5	92.4	3.6	5.7

Table 7: MGTD-head four-quadrant results (%): recall for human (H) subsets, false-positive rate for AI subsets. s = sarcastic, n = non-sarcastic.

The single-head MGTD controls remain within single-digit percentage points of the dual-head models in every quadrant. No analogous extreme class-conditional error appears on the MGTD side, consistent with §5.3.

No comparable subgroup increase appears in the qwen3.5:9b replacement set: AI × non-sarcastic

false-positive rates are 0.9%, 1.3%, and 0.5% for plain, interaction projection, and RCNN; AI $\times$ sarcastic rates are at most 0.4%; and overall MGTD F1 scores are 0.952, 0.917, and 0.966.

C.3 Blinded-evaluation protocol, inter-judge agreement, and group-level tests (§6). Each sample is submitted separately to each of the three judge APIs using a single prompt (May 2026; temperature = 0.3; GPT-5-nano uses its default decoding because it does not support this parameter). The prompt contains only the input and evaluated text, not the ground-truth source label. It asks for source inference followed by 1–5 ratings of sarcasm and human-likeness. We therefore refer to these ratings as source-label-blinded. Base and SFT outputs use the same prompt and are evaluated separately.

Pairwise instance-level agreement is low to moderate. Spearman ρ ranges from 0.20 to 0.38 for sarcasm (all $p < 10^{-6}$) and from 0.03 to 0.23 for human-likeness (one pair is not significant). The conclusion in §6 therefore rests on consistent condition-level directions, not agreement in instance rankings.

Wilcoxon signed-rank tests with Benjamini–Hochberg FDR correction (Benjamini and Hochberg, 1995) show that SFT receives lower sarcasm ratings than the base model in five of six language $\times$ judge groups. The pooled correlations in Table 8 use the three-judge mean as a descriptive aggregate; because inter-judge agreement is limited, the judge-specific results remain necessary. Four instances each have one unparseable rating and retain two valid judge ratings; the aggregate uses the mean of the available ratings and therefore retains all 600 outputs.

Subset	n	Sarc. ρ	Human-like ρ
Overall	600	+0.136*	−0.212*
Chinese	360	−0.020	−0.218*
English	240	+0.338*	−0.270*

Table 8: Spearman correlations between detector probabilities and LLM-judge ratings (RCNN detector, $n = 600$ pooled over base and SFT): Sarc. ρ compares $P(\text{sarcastic})$ with sarcasm ratings, and Human-like ρ compares $P(\text{human})$ with human-likeness ratings. * significant; the other entries are not significant.

C.4 Paired tests between SFT and human references (supplementary). These tests are performed on the Chinese subset of the generation-

Judge / lang.	n	Base	SFT	adj. p
DeepSeek-V3 / en	120	4.025	3.292	<.0001
DeepSeek-V3 / zh	180	4.044	3.878	.0008
Gemini-2.5-FL / en	120	3.992	3.750	.0008
Gemini-2.5-FL / zh	180	4.028	3.994	.2008*
GPT-5-nano / en	119	4.084	3.681	<.0001
GPT-5-nano / zh	180	3.911	3.517	<.0001

Table 9: Group-level tests for blinded sarcasm judgments (Wilcoxon + BH-FDR). * not significant. GPT-5-nano lacks one English base rating (C.3: $n = 119$) and three Chinese reference ratings (affecting C.4, not the C.3 Chinese row); tests use pairwise deletion.

evaluation set ($n = 180$; three GPT-5-nano ratings are missing, leaving 177). The comparisons provide judge- and dimension-specific calibration context: for DeepSeek-V3, SFT is significantly below the reference on human-likeness (-0.161 , $p = 0.0025$), whereas for GPT-5-nano, SFT is significantly above the reference on sarcasm ($+0.186$, $p = 0.0114$). The remaining four comparisons do not reach significance at the reported threshold. Taken together, the results indicate that proximity to the human reference depends on the judge and evaluated dimension.

C.5 Plain-detector control. With the plain detector, the Chinese sarcasm probability changes from 0.765 to 0.763 (-0.001), the English sarcasm probability changes from 0.843 to 0.659 (-0.184), and predicted human-source probability rises by $+0.760$ in Chinese and $+0.705$ in English. The two detectors therefore agree on the rise in predicted human-source probability, while the sarcasm direction depends on the detector.

Correlations between detector probabilities and corresponding ratings are $+0.142$ for sarcasm and -0.214 for human-likeness overall; -0.037 (not significant) and -0.209 in Chinese; and $+0.342$ and -0.283 in English. Each differs from the RCNN-detector value by at most 0.02. Negative associations between predicted human-source probabilities and human-likeness ratings are similar across the two detectors, whereas the condition-level sarcasm-probability change is architecture-dependent, especially in Chinese.

Within-condition correlations (RCNN detector). For predicted human-source probability and human-likeness ratings, the base condition has $\rho = +0.03$ ($n = 300$, not significant; Chinese $+0.10$, English -0.09). The SFT condition has

$\rho = -0.12$ ($p = 0.039$; Chinese -0.19 , $p = 0.009$; English -0.12 , not significant). The pooled value of -0.212 is driven mainly by the opposing condition-level changes: SFT raises predicted human-source probability by about 0.67 while lowering the mean blinded rating. There is no significant positive detector–judge alignment on this dimension within conditions.

For sarcasm, within-condition correlations are $+0.05$ for Chinese base (not significant), $+0.15$ for Chinese SFT ($p = 0.048$), $+0.41$ for English base, and $+0.37$ for English SFT (both English values significant). Chinese alignment is therefore at most weakly positive, consistent with its near-zero pooled result.

C.6 Detector probabilities before and after SFT across the two task–language settings. The RCNN detector assigns the following mean probabilities to base and SFT outputs:

Lang.	Cond.	$P(\text{sarcastic})$	$P(\text{human})$
Chinese	Base $\rightarrow$ SFT	0.780 $\rightarrow$ 0.895	0.289 $\rightarrow$ 0.969
English	Base $\rightarrow$ SFT	0.854 $\rightarrow$ 0.730	0.198 $\rightarrow$ 0.864

Table 10: Mean detector probabilities before and after SFT (RCNN detector).

Sarcasm probabilities move in opposite directions in the Chinese comment-generation and English rewriting settings: $+0.115$ and -0.124 , respectively. GenTrain itself is balanced at 1,214 instances per language; the evaluation settings instead differ in task, prompt format, source data, and sample size, all of which may contribute to the contrast. Predicted human-source probabilities rise sharply in both settings.

The mean LLM ratings for human references are 3.77 for sarcasm and 4.17 for human-likeness; the corresponding values are 4.01 and 4.24 for the base model and 3.71 and 4.08 for SFT. From the base model to SFT, the mean human-likeness rating decreases even as predicted human-source probability increases, as analyzed in §6.

C.7 Conditional test-instance uncertainty: confidence intervals, paired tests, and subset F1. Conditional on each fixed final model, we quantify test-instance uncertainty with Wilson 95% intervals for quadrant rates and 95% percentile intervals from 10,000 instance-level bootstrap samples for F1. These intervals do not include uncertainty from training, generator sampling, corpus

construction, or human–AI pairing. They complement the between-run standard deviations in Table 2 and Table 1. All values use the final model for each architecture and follow the reporting convention of Table 3.

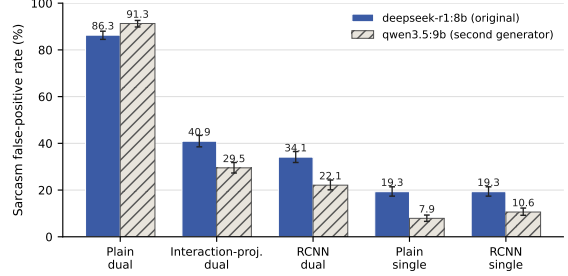


Figure 2: Sarcasm false-positive rates on the AI $\times$ non-sarcastic subset for five detectors and two generators. Error bars are Wilson 95% intervals conditional on each fixed model.

Both matched dual–single contrasts retain the same direction on the Qwen3.5-generated AI texts, although their magnitudes differ.

Model	DeepSeek	Qwen3.5
Plain dual	86.3 [84.5, 88.0]	91.3 [89.8, 92.6]
Inter.-proj. dual	40.9 [38.5, 43.4]	29.5 [27.3, 31.8]
RCNN dual	34.1 [31.8, 36.5]	22.1 [20.1, 24.3]
Plain single	19.3 [17.4, 21.4]	7.9 [6.6, 9.3]
RCNN single	19.3 [17.4, 21.4]	10.6 [9.2, 12.3]

Table 11: Wilson 95% intervals for the sarcasm false-positive rate on the AI $\times$ non-sarcastic subset (two generators; %).

Paired McNemar tests use predictions on the same DeepSeek-generated AI $\times$ non-sarcastic texts. The discordant-pair counts are 1,021 / 7 for plain dual vs. plain single, 239 / 15 for RCNN dual vs. RCNN single, 805 / 15 for plain dual vs. RCNN dual, and 203 / 100 for interaction projection vs. RCNN dual (all $p < .001$). The discordant counts and absolute rate differences provide the substantive effect-size information.

In both architectures, the dual–single difference therefore has the same direction in Chinese and English. Language remains coupled with corpus, genre, and annotation setting, so these results provide same-direction evidence across the two task–language settings rather than an isolated language effect.

The final RCNN dual-head model’s MGTD F1 is 0.938 [0.931, 0.945] on the full set, 0.961 [0.952, 0.969] in Chinese, and 0.923 [0.912, 0.933] in En-

Model	Lang.	<i>n</i>	Dual FP; rate [CI]	Single FP; rate [CI]	Δ pp
Plain	en	1,174	987; 84.1 [81.9, 86.1]	242; 20.6 [18.4, 23.0]	+63.5
Plain	zh	339	319; 94.1 [91.1, 96.2]	50; 14.7 [11.4, 18.9]	+79.4
RCNN	en	1,174	445; 37.9 [35.2, 40.7]	247; 21.0 [18.8, 23.5]	+16.9
RCNN	zh	339	71; 20.9 [17.0, 25.6]	45; 13.3 [10.1, 17.3]	+7.7

Table 12: Language-stratified sarcasm false-positive rates on AI texts prompted to be non-sarcastic (%; Wilson 95% intervals). Dual FP and Single FP give the number of false positives under the dual- and single-head models. Exact paired McNemar tests give $p < .001$ for both plain comparisons and RCNN English, and $p = 2.4 \times 10^{-5}$ for RCNN Chinese.

Model	Full set	Chinese	English	Human subset	Human $\times$ zh	Human $\times$ en
Plain dual-head	0.556	0.746	0.312	0.586	0.710	0.374
Interaction-projection dual-head	0.643	0.781	0.420	0.593	0.739	0.366
RCNN dual-head	0.663 [.646, .680]	0.819 [.802, .836]	0.429 [.399, .458]	0.629 [.605, .653]	0.774 [.748, .798]	0.399 [.355, .440]
Plain single-head	0.661	0.750	0.514	0.612	0.719	0.434
RCNN single-head	0.664	0.758	0.511	0.603	0.709	0.434

Table 13: Per-language and human-subset sarcasm F1 (final models; bootstrap 95% intervals included for RCNN).

glish. Because the human subset contains no AI-labeled negatives, a source-class F1 is not separately reported for this subset; Table 7 reports human recall directly.

Chinese sarcasm F1 is higher than English sarcasm F1 for the RCNN model (0.819 vs. 0.429). This contrast also reflects differences in prevalence, genre, and annotation procedure: sarcasm prevalence is 63.8% in Chinese but 14.2% in English, and the English human side is dominated by author-labeled intended-sarcasm tweets from iSarcasmEval whereas the Chinese side uses topic-comment-style ToSarcasm. The positive-class F1 values should therefore not be interpreted as an isolated language effect.

The separately retrained final RCNN model’s overall sarcasm F1 is 0.663, compared with 0.629 on human texts alone, because the overall score also includes AI texts. The five-run mean in Table 1 is a different estimate. Within the human subset, dual–single differences are below 0.03 and vary in direction. The negative-transfer effect is therefore concentrated in the AI-side quadrants, consistent with §5.3.

False-positive rates use a threshold of 0.5. For the plain dual-head model on AI $\times$ non-sarcastic text, the median sarcasm probability is 0.767 with an interquartile range of 0.635–0.839. The distribution therefore lies mostly above the threshold rather than placing only a few cases near it. Single-head medians are 0.204 for the plain model and 0.069 for RCNN.

AUC values within AI texts are similar across models (0.82–0.88). Joint training thus raises sar-

casm scores while largely preserving their internal ordering. This pattern is consistent with a source-related component being added to the sarcasm signal.

D Illustrative Cases of Detector–LLM Score Misalignment

Illustrative cases were manually selected after inspecting the paired table of RCNN-detector scores and ratings from the three LLMs to show large disagreements; no formal ranking or top- k selection was performed. Detector scores range from 0 to 1, and LLM scores are averaged across the three evaluators on a 1–5 scale; long texts are truncated. Chinese examples retain the original model input and provide a pinyin transliteration and an English translation in brackets.

The three groups show, respectively, high predicted human-source probabilities with low LLM human-likeness ratings, high predicted sarcasm probabilities with low LLM sarcasm ratings, and the reverse sarcasm disagreement. In the first human-likeness case, AI-generated text with obvious chain-of-thought traces is nevertheless assigned predicted human-source probability 1.00 by the MGTD head, corresponding to the disagreement discussed in §6.

Together, these selected cases motivate the hypothesis that surface markers such as mock-praise formulas, exclamation marks, and colloquial short sentences contribute to some detector decisions, while semantic reversal contributes to some LLM-judge ratings. They define candidate features for systematic tests of feature prevalence and effects

L	S	$P_{\text{hum}}/\text{LLM}$	Text (excerpt)
zh	SFT	1.00 / 2.67	“好，我需要针对三星回怼华为的新闻生成一条讽刺评论。首先……” [hǎo, wǒ xūyào zhēnduì sān xīng huíduì huáwéi de xīnwén shēngchéng yī tiáo fěngcì pínglùn. shǒuxiān...] [“Okay, I need to generate a sarcastic comment on the news of Samsung hitting back at Huawei. First...”] (chain-of-thought leakage)
zh	SFT	1.00 / 2.33	“跑分竞赛：参数焦虑：跑分焦虑：水货：哈哈哈” [pǎofēn jìngsài: cānshù jiāolǜ: pǎofēn jiāolǜ: shuǐhuò: hāhāhā] [“Benchmark race: spec anxiety: benchmark anxiety: grey-market goods: hahaha”] (degenerate output)
en	SFT	1.00 / 2.67	“Same as me, I’m not sure” (vacuous short sentence)

Table 14: Predicted human-source probability / human-likeness misalignment cases. L = language; S = model condition; the third column gives the detector probability / mean LLM rating, with $P_{\text{hum}} = P(\text{human})$.

L	S	$P_{\text{sar}}/\text{LLM}$	Text (excerpt)
zh	SFT	1.00 / 2.67	“三星：我才是第一！你看看我这摄像头总分，109+96，112+89，我赢了！” [Sān xīng: wǒ cái shì dì yī! nǐ kànkàn wǒ zhè shèxiàngtóu zǒngfēn, 109+96, 112+89, wǒ yíng le!] [“Samsung: I’m the real number one! Look at my camera total scores, 109+96, 112+89, I win!”] (numbers repeated without sarcastic transformation)
zh	SFT	0.99 / 2.67	“恭喜恭喜，美国牛牛牛，继续加油” [gōngxǐ gōngxǐ, Měiguó niú niú niú, jìxù jiāyóu] [“Congratulations, congratulations, America is awesome awesome awesome, keep it up”] (formulaic mock-praise)
en	SFT	0.93 / 2.67	“2 hours left to tip it over, I’m going to the toilet”

Table 15: Detector-high / LLM-judge-low sarcasm cases. L = language; S = model condition; the third column gives the detector probability / mean LLM rating, with $P_{\text{sar}} = P(\text{sarcastic})$.

L	S	$P_{\text{sar}}/\text{LLM}$	Text (excerpt)
zh	Base	0.14 / 4.00	“特朗普承诺 120 亿补贴像撒大饼，农民收到的却是缩水成硬币的承诺……” [Tèlǎngpǔ chéngnuò 120 yì bǔtiē xiàng sā dàbǐng, nóngmín shōudào de què shì suōshuǐ chéng yìngbì de chéngnuò...] [“Trump promised a 12-billion subsidy as if tossing out big pancakes, yet what farmers receive are promises shrunk into coins...”] (metaphorical irony)
zh	Base	0.24 / 4.00	“特斯拉：价格跳水后又跳水，消费者：钱包跳水后又跳水。” [Tèsīlā: jiàgé tiàoshuǐ hòu yòu tiàoshuǐ, xiāofèizhě: qiánbāo tiàoshuǐ hòu yòu tiàoshuǐ.] [“Tesla: prices dive and then dive again; consumers: wallets dive and then dive again.”] (parallel-structure irony)
en	Base	0.12 / 4.00	“Thank you so much for this *beloved* review, Mandy! I’m *overjoyed* you enjoyed it” (asterisk emphasis)

Table 16: Detector-low / LLM-judge-high sarcasm cases. L = language; S = model condition; the third column gives the detector probability / mean LLM rating, with $P_{\text{sar}} = P(\text{sarcastic})$.

and provide qualitative examples of the near-zero Chinese sarcasm correlation in §6.

E Representation-Level Analysis of Source-Conditioned Error Allocation

This appendix records the complete protocol, data, and self-checks for the representation-level tests in §5.3.2. The conclusions are interpreted as described in §5.3.2.

E.1 Setup. All analyses use the frozen plain dual-head model that produces the 86.3% result reported in the main text. We use split-sample evaluation: one half is used to estimate the direction, and the other is used to compute metrics. The appendix baseline is therefore 84.8% on the evaluation half rather than 86.3% on the full set.

Implementation checks pass. Manually computed logits match model outputs with a maximum error of 1.4×10^{-6} , and the erasure-projection residual is about 10^{-6} . LEACE satisfies its defining property: after erasure, the fitting-half class-mean difference has a relative residual of 7×10^{-6} , and an in-sample linear probe is at chance (AUC 0.514). The source-probe AUC is 0.986 with true labels and 0.512 under source-label permutation.

E.2 Protocol. The linear probe is a logistic-regression classifier evaluated with five-fold cross-validation. A permuted-label control retrain the same probe after shuffling source labels (Hewitt and Liang, 2019), and correlation analyses are repeated within text-length tertiles. For alignment, Σ is the Ledoit–Wolf shrinkage covariance estimator (Ledoit and Wolf, 2004) fitted to all pooled Dual-

Test representations, with eigenvalues clipped below 10^{-8} ; w denotes a task-head weight difference or probe direction. Whitenened alignment is $\cos(\Sigma^{1/2}w_1, \Sigma^{1/2}w_2)$.

Erasure tests cover one direction, INLP subspaces ($k = 1-16$; Ravfogel et al., 2020), and LEACE (Belrose et al., 2023). Controls use ten sampled random whitened directions and directions estimated from permuted labels. We construct source-direction-orthogonal PC1 (srcORTH-PC1) by projecting the fitting-half representations onto the orthogonal complement of the estimated linear source direction and then extracting the maximum-variance direction in that projected space. Erasure removes each representation’s scalar projection onto this direction. This operation leaves the measured linear source decodability unchanged, although orthogonality to one estimated direction does not imply independence from all multidimensional or nonlinear source information.

E.3 Diagnostic evidence. Source is highly linearly decodable: AUC is 0.986 overall, 0.990 on sarcastic texts, and 0.985 on non-sarcastic texts. The permuted-label control gives 0.512. The single-head model also yields an AUC of 0.974, showing that source decodability is a background property shared by both models. On $\text{AI} \times \text{non-sarcastic}$ texts, the MGTD-head AI logit and sarcasm-head logit correlate at $\rho = 0.519$; correlations by text-length tertile are 0.44, 0.56, and 0.63.

The sarcasm and source directions are nearly orthogonal in the original geometry ($\cos \approx -0.03$) but moderately aligned after whitening. The absolute cosine is 0.41 using the probe direction and 0.49 using the MGTD-head direction. These results provide correlational evidence consistent with the shortcut hypothesis and motivate the intervention tests below.

E.4 Intervention evidence. $\text{gap} = \text{AI} \times \text{non-sarcastic false-positive rate} - \text{human} \times \text{non-sarcastic false-positive rate}$; source AUC measures the performance of a probe retrained after erasure.

Single-direction and low-dimensional INLP erasure produce little change. LEACE reduces measured linear source decodability and the source-specific gap from 52.0 to 19.1; the ten sampled random-direction controls show no comparable gap change.

The orthogonal high-variance control qualifies this interpretation. It leaves the measured linear

Erasure scheme	AI×n FP	H×n FP	Gap	AUC
None (baseline)	84.8	32.8	52.0	0.982
Single dir. (head / probe / random)	84.8–86.2	—	—	≈ 0.98
INLP ($k = 16$)	89.8	—	—	0.974
LEACE (linear)	68.7	49.5	19.1	0.589
Random	84.9	32.7	—	0.982
whitened ×10				
srcORTH-PC1	66.7	54.1	12.6	0.982

Table 17: Intervention evidence. $\text{H} \times \text{n}$ = human $\times$ non-sarcastic; $\text{AI} \times \text{n}$ = AI $\times$ non-sarcastic; FP in %; AUC is the source-probe AUC after erasure. Bold marks the two diagnostic interventions discussed in §5.3.2, not best performance.

source decodability intact but changes the AI-side rate to 66.7 and the source gap to 12.6, an effect similar to LEACE. Together with the random-direction controls, this result shows that predictions are sensitive to a non-random high-variance direction constructed after removing its projection onto the estimated source direction. The direction’s semantic content and causal role therefore remain focused targets for further analysis.

Under LEACE, the unweighted mean false-positive rate across the two non-sarcastic source groups is almost unchanged ($58.8 \rightarrow 59.1$). This result shows that the affected directions shape how non-sarcastic errors are allocated between AI and human texts.